



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (PRELIMINARY COURSE)

2010 Assessment Task 2

General instructions

- Working time – 1 period.
- Commence each new question on a new sheet.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.

Class (please ✓)

☐ 11M2A – Mr Lowe

NAME: **# PAGES USED:**

Marker's use only.

QUESTION	1	2	3	4	5	Total	%
MARKS	$\overline{6}$	$\overline{15}$	$\overline{9}$	$\overline{12}$	$\overline{11}$	$\overline{53}$	

Question 1	(6 Marks)	Commence a NEW page.	Marks
(a)	Find the value of $\tan 26^\circ$ correct to 3 decimal places.		1
(b)	Find the size of the acute angle A in degrees and minutes (to the nearest minute) for $\sin A = \frac{5}{8}$.		1
(c)	Why would an error be generated when $\cos^{-1}\left(\frac{8.4}{7.6}\right)$ is entered on the calculator?		1
(d)	Find the exact value of the following expressions:		
	i. $\sin 45^\circ$.		1
	ii. $\cos 30^\circ$.		1
	iii. $\cot 210^\circ$.		1

Question 2	(15 Marks)	Commence a NEW page.	Marks
(a)	Solve for $0 \leq \theta < 360^\circ$:		
	i. $\sin \theta = \frac{1}{2}$.		2
	ii. $\cos 2\theta = \frac{\sqrt{3}}{2}$.		3
	iii. $\tan^2 \theta = 1$.		3
	iv. $2\cos^2 \theta - \cos \theta = 0$.		3
(b)	For the curve $y = 2\sin 3\theta$,		
	i. State the period.		1
	ii. State the amplitude.		1
	iii. Sketch the curve for $0^\circ \leq \theta \leq 360^\circ$.		2

Question 3 (9 Marks)

Commence a NEW page.

Marks

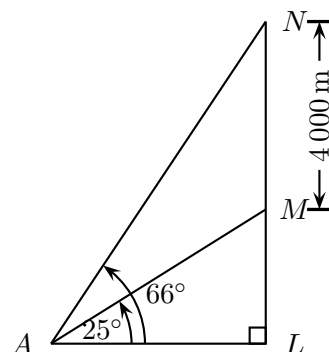
- (a) For $A(2, 4)$, $B(5, 6)$, find the
- gradient of AB . 1
 - midpoint of AB . 2
 - distance between the points. 1
 - equation of AB . 2
- (b) Given the line $2x - 3y = 4$,
- Express in gradient-intercept form. 1
 - State the gradient. 1
 - Find the angle the line makes with the positive x axis, correct to the nearest minute. 1

Question 4 (12 Marks)

Commence a NEW page.

Marks

- (a) In $\triangle ABC$, $\sin B = \frac{4}{5}$, $a = 6$ and $b = 9$. Find $\sin A$ as the simplest exact value. 2
- (b) Two sides of a triangular field are 60 m and 50 m with the included angle being 140° . By sketching a diagram, calculate correct to 3 decimal places:
- the length of the third side. 2
 - the area of the field. 2
- (c) A rocket launched vertically from L is observed from A . Soon after the launch when at position M its angle of elevation is 25° . After it climbs 4 000 m to position N , its angle of elevation is 66° .



NOT TO SCALE

- Find $\angle ANL$ and $\angle NAM$. 2
- Use the sine rule to find AM . 2
- Find how far the observer is from the launch pad correct to the nearest metre. 2

Question 5	(11 Marks)	Commence a NEW page.	Marks
(a)	Simplify $\cos^3 \theta + \cos \theta \sin^2 \theta$.		2
(b)	If $\tan \alpha = -\frac{5}{12}$ and $0^\circ \leq \alpha \leq 360^\circ$, find the exact value of $\sin \alpha$.		2
(c)	Prove $\frac{1}{1 - \cos \theta} + \frac{1}{1 + \cos \theta} = 2 \operatorname{cosec}^2 \theta$.		3
(d)	Two cars leave point A at the same time. One car averages 80 km/h along a straight road in the direction 025°T . The other car averages 90 km/h along a straight road in the direction 135°T . How far apart (to the nearest kilometre) are they after 3 hours?		3

End of paper.

Suggested Solutions

Question 1

(a) (1 mark)

$$\tan 26^\circ = 0.488 \text{ (3 d.p.)}$$

(b) (1 mark)

$$A = \sin^{-1} \frac{5}{8} = 38^\circ 41'$$

(c) (1 mark)

As $-1 \leq \cos \theta \leq 1$, and from the expression, $\cos \theta > 1$, which is out of range.

(d) i. (1 mark)

$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

ii. (1 mark)

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

iii. (1 mark)

$$\begin{aligned} \cot 210^\circ &= \frac{1}{\tan(180^\circ + 30^\circ)} \\ &= \frac{1}{\tan 30^\circ} = \frac{1}{\frac{1}{\sqrt{3}}} \\ &= \sqrt{3} \end{aligned}$$

Question 2

(a) i. (2 marks)

$$\begin{aligned} \sin \theta &= \frac{1}{2} \\ \theta &= 30^\circ, 150^\circ \end{aligned}$$

ii. (3 marks)

$$\begin{aligned} 0^\circ &\leq \theta < 360^\circ \\ \therefore 0^\circ &\leq 2\theta < 720^\circ \\ \cos 2\theta &= \frac{\sqrt{3}}{2} \\ 2\theta &= 30^\circ, 330^\circ, 390^\circ, 690^\circ \\ \therefore \theta &= 15^\circ, 165^\circ, 195^\circ, 345^\circ \end{aligned}$$

iii. (3 marks)

$$\begin{aligned} \tan^2 \theta &= 1 \\ \tan \theta &= \pm 1 \\ \therefore \theta &= 45^\circ, 135^\circ, 225^\circ, 315^\circ \end{aligned}$$

iv. (3 marks)

$$\begin{aligned} 2 \cos^2 \theta - \cos \theta &= 0 \\ \cos \theta (2 \cos \theta - 1) &= 0 \\ \cos \theta = 0 &\quad \left| \quad \cos \theta = \frac{1}{2} \right. \\ \therefore \theta = 90^\circ, 270^\circ &\quad \left| \quad \therefore \theta = 60^\circ, 300^\circ \right. \\ \therefore \theta = 60^\circ, 90^\circ, 270^\circ, 300^\circ \end{aligned}$$

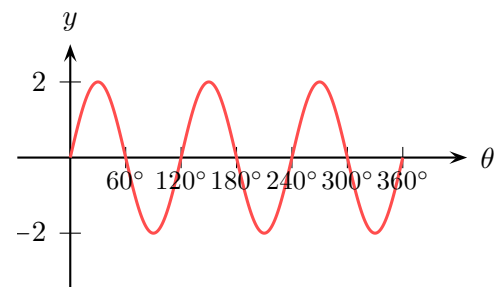
(b) i. (1 mark)

$$T = \frac{360^\circ}{3} = 120^\circ$$

ii. (1 mark)

$$a = 2$$

iii. (2 marks)



Question 3

(a) i. (1 mark)

$$\begin{aligned} A(2, 4) \quad B(5, 6) \\ m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{6 - 4}{5 - 2} = \frac{2}{3} \end{aligned}$$

ii. (2 marks)

$$\begin{aligned}
 (x, y) &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \\
 &= \left(\frac{5 + 2}{2}, \frac{6 + 4}{2} \right) \\
 &= \left(\frac{7}{2}, 5 \right)
 \end{aligned}$$

iii. (1 mark)

$$\begin{aligned}
 d &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\
 &= \sqrt{(5 - 2)^2 + (6 - 4)^2} \\
 &= \sqrt{3^2 + 2^2} = \sqrt{13}
 \end{aligned}$$

iv. (2 marks)

$$\begin{aligned}
 \frac{y - y_1}{x - x_1} &= m \\
 \frac{y - 6}{x - 5} &= \frac{2}{3} \\
 \times (x - 5) &\quad \times (x - 5) \\
 y - 6 &= \frac{2}{3}x - \frac{10}{3} \\
 y &= \frac{2}{3}x + \frac{8}{3}
 \end{aligned}$$

(b) i. (1 mark)

$$\begin{aligned}
 2x - 3y &= 4 \\
 -3y &= 4 - 2x \\
 3y &= 2x - 4 \\
 y &= \frac{2}{3}x - \frac{4}{3}
 \end{aligned}$$

ii. (1 mark)

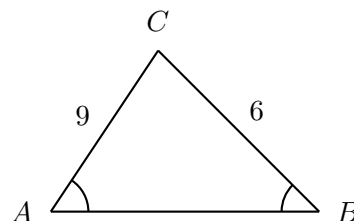
$$m = \frac{2}{3}$$

iii. (1 mark)

$$\begin{aligned}
 m &= \tan \theta \\
 \tan \theta &= \frac{2}{3} \\
 \therefore \theta &= \tan^{-1} \frac{2}{3} \\
 &= 33.69^\circ = 33^\circ 41'
 \end{aligned}$$

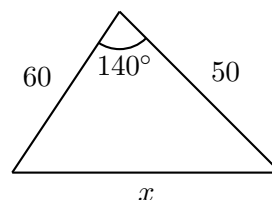
Question 4

(a) (2 marks)



$$\begin{aligned}
 \frac{\sin A}{6} &= \frac{\sin B}{9} \\
 \sin A &= \frac{6}{9} \times \sin B \\
 &= \frac{2}{3} \times \frac{4}{5} \\
 &= \frac{8}{15}
 \end{aligned}$$

(b) i. (2 marks)



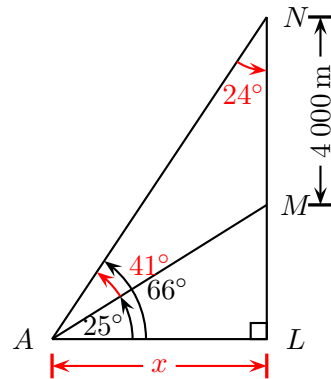
Using the cosine rule,

$$\begin{aligned}
 x^2 &= 60^2 + 50^2 - 2 \times 60 \times 50 \times \cos 140^\circ \\
 &= 10696.267 \dots \\
 x &= 103.423 \text{ m}
 \end{aligned}$$

ii. (2 marks)

$$\begin{aligned}
 A &= \frac{1}{2}ab \sin C \\
 &= \frac{1}{2} \times 60 \times 50 \times \sin 140^\circ \\
 &= 964.181 \text{ m}^2
 \end{aligned}$$

(c) i. (2 marks)



$$\angle ANL = 90^\circ - 66^\circ = 24^\circ$$

$$\angle NAM = 66^\circ - 25^\circ = 41^\circ$$

ii. (2 marks)

$$\begin{aligned} \frac{AM}{\sin 24^\circ} &= \frac{4000}{\sin 41^\circ} \\ AM &= \frac{4000 \sin 24^\circ}{\sin 41^\circ} \\ &= 2479.88 \text{ m} \end{aligned}$$

iii. (2 marks)

$$\begin{aligned} \frac{x}{AM} &= \cos 25^\circ \\ x &= AM \cos 25^\circ \\ &= \frac{4000 \sin 24^\circ}{\sin 41^\circ} \times \cos 25^\circ \\ &= 2247.53 \text{ m} \\ &= 2248 \text{ m (nearest metre)} \end{aligned}$$

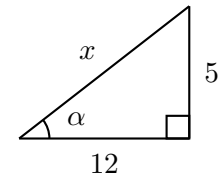
Question 5

(a) (2 marks)

$$\begin{aligned} &\cos^3 \theta + \cos \theta \sin^2 \theta \\ &= \cos \theta (\cos^2 \theta + \sin^2 \theta) \\ &= \cos \theta \end{aligned}$$

(b) (3 marks)

- ✓ [1] for locating α in the 2nd or 4th quadrants
- ✓ [1] for finding correctly the hypotenuse in the corresponding $\triangle$
- ✓ [1] for final correct answer

Alternatively, $[-1]$ for each mistake.

$$\tan \alpha = -\frac{5}{12}$$

 $\therefore \alpha$ is in the second or fourth quadrants

$$\therefore x = \sqrt{12^2 + 5^2} = 13$$

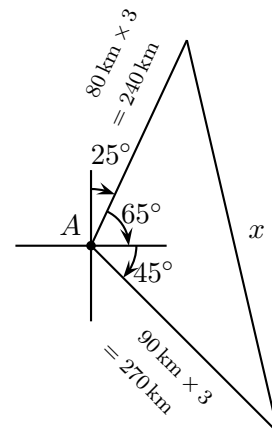
$$\therefore \sin \alpha = \pm \frac{5}{13}$$

(c) (3 marks)

$$\begin{aligned} &\frac{1}{1 - \cos \theta} + \frac{1}{1 + \cos \theta} \\ &= \frac{(1 + \cos \theta) + (1 - \cos \theta)}{(1 - \cos \theta)(1 + \cos \theta)} \\ &= \frac{2}{1 - \cos^2 \theta} \\ &= \frac{2}{\sin^2 \theta} \\ &= 2 \operatorname{cosec}^2 \theta \end{aligned}$$

(d) (3 marks)

- ✓ [1] for diagram
- ✓ [1] for attempt to use cosine rule
- ✓ [1] for final correct answer



Apply the cosine rule to the triangle,

$$\begin{aligned} x^2 &= 240^2 + 270^2 - 2 \times 240 \times 270 \times \cos 110^\circ \\ &= 174\,825.81 \dots \end{aligned}$$

$$x = 418.12 \text{ km}$$